Quantum-Safe MIME Email System

using the McEliece Key Encapsulation Mechanism

Project Report

Post-Quantum Cryptography — Applied Implementation

"Replacing RSA/ECDH with code-based cryptography to build an asynchronous, store-and-forward email system resistant to quantum attacks."

Contents

Abstract

The rapid advancement of quantum computing poses a direct threat to public-key cryptosystems that underpin modern secure communication — including RSA and elliptic-curve Diffie-Hellman (ECDH), both of which can be broken in polynomial time by Shor's algorithm. This project presents a *quantum-safe*, asynchronous, store-and-forward email system built on top of the MIME protocol. The system replaces conventional RSA-based key exchange with the **McEliece Key Encapsulation Mechanism (KEM)**, a code-based cryptosystem whose security rests on the NP-complete syndrome decoding problem — believed to be intractable for both classical and quantum computers.

Every client–server connection executes a fresh McEliece KEM handshake, deriving an ephemeral 256-bit AES session key. Stored emails are re-encrypted under an independent, per-message random storage key, providing both forward secrecy and storage-level isolation. The implementation supports structured MIME messages including multi-part attachments, multiple users, and asynchronous delivery (sender and recipient need not be online simultaneously). Performance benchmarks for five security parameter sets are reported, ranging from $\sim$1.8 s handshake time at the recommended medium level to $\sim$36 min at the maximum-security level.

1 Introduction

Electronic mail remains one of the most widely used forms of digital communication. Conventional secure email standards such as S/MIME and PGP rely on RSA or ECDH for key exchange, algorithms that are vulnerable to Shor's quantum algorithm [2]. While large-scale quantum computers capable of running Shor's algorithm at cryptographically relevant key sizes do not yet exist, the threat of *harvest-now, decrypt-later* attacks — adversaries collecting encrypted traffic today with the intent to decrypt it once quantum hardware matures — is already real.

This project addresses the problem by substituting the RSA KEM with the **McEliece cryptosystem** [1], one of the oldest and most studied post-quantum proposals, and a NIST Post-Quantum Cryptography candidate [3]. The result is a complete, working email system with the following properties:

- **Quantum-resistant key exchange** via McEliece KEM.
- **Symmetric encryption** using AES-256-CBC for bulk data.
- **Forward secrecy** through ephemeral per-connection KEM handshakes.
- **Storage isolation** via independent per-email random keys.
- **Asynchronous delivery** — sender and recipient operate independently; emails are stored on the server and fetched later.
- **Full MIME support** including multi-part messages and file attachments (RFC 2045–2049 [4]).

2 Background

2.1 The Quantum Threat to Classical Cryptography

RSA and ECDH owe their security to the computational hardness of integer factorisation and the discrete logarithm problem, respectively. Shor (1994) showed that both can be solved in polynomial time on a quantum computer. Even assuming quantum hardware with sufficient coherence and qubit count remains years away, the harvest-now-decrypt-later attack vector makes migration to post-quantum algorithms urgent for long-lived sensitive communications.

Grover's algorithm provides a quadratic speedup for unstructured search, which reduces the effective key length of symmetric ciphers by half (e.g. AES-256 degrades to $\approx$ 128-bit security). AES-256 is therefore still considered safe against quantum adversaries, making it the natural choice for bulk encryption in a post-quantum system.

2.2 The McEliece Cryptosystem

Introduced by Robert J. McEliece in 1978, the McEliece cryptosystem is a public-key scheme based on *error-correcting codes*. Its security derives from the hardness of decoding a random linear code — the *syndrome decoding problem* — which is NP-complete and has resisted more than four decades of cryptanalysis.

2.2.1 Key Components

Generator matrix G
An $k \times n$ binary matrix from a chosen error-correcting code (e.g. Goppa codes) capable of correcting up to t errors.

Scrambling matrix S
A random invertible $k \times k$ binary matrix.

Permutation matrix P
A random $n \times n$ permutation matrix.

Public key
$\hat{G} = S \cdot G \cdot P$, which looks like a random linear code to an adversary.

Private key
(S, G, P) — the structured decomposition.

2.2.2 Key Encapsulation Mechanism (KEM) Construction

In the KEM variant used in this project, the McEliece scheme is adapted as follows:

1. **Key generation (server):** Generate Goppa-code parameters (q, k, t); compute public matrix $G_{\bar{\omega}}$.
2. **Encapsulation (client):** Choose a random message m_ω; compute ciphertext $c = m_\omega \hat{G} + e$ where e is a random error vector of weight $\leq t$; derive shared secret via HKDF.
3. **Decapsulation (server):** Use private key to correct errors in c and recover m_ω; derive the same shared secret via HKDF.

The shared secret (32 bytes) is then used directly as an AES-256-CBC key for that connection.

2.3 MIME Protocol Overview

MIME (Multipurpose Internet Mail Extensions), defined in RFC 2045–2049, extends the basic Internet Message Format (RFC 2822) to support:

- Multiple content types (text, HTML, binary).
- Multi-part messages with distinct body sections.
- Base64-encoded binary attachments.
- Internationalised headers.

The system implements a subset of the MIME standard sufficient for plain-text emails with arbitrary file attachments, without relying on SMTP for transport.

3 System Architecture

3.1 High-Level Design

The system follows a *client–server* architecture with a *store-and-forward* model identical in spirit to classical SMTP/IMAP, but with every network interaction guarded by a fresh McEliece KEM handshake.

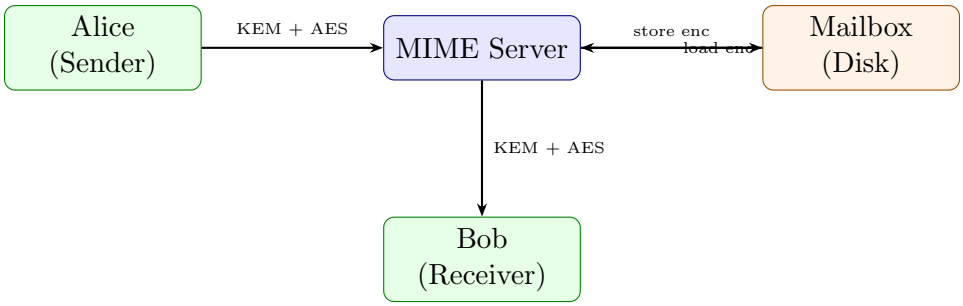

Figure 1: High-level system architecture: every arrow represents an AES-encrypted channel established via McEliece KEM handshake. Emails persist on disk between send and receive operations.

3.2 Module Breakdown

Table 1 summarises the principal source files and their responsibilities.

Table 1: Source modules and roles

Module	Responsibility	LoC (approx
`mime_server_with_kem.py`	TCP server; KEM handshake per connection; handles SEND/RECV/LIST commands	~30
`mime_client_with_kem.py`	CLI client; builds MIME messages; encrypts/decrypts payloads	~40
`mailbox_manager.py`	File-based mailbox; stores metadata JSON, encrypted content, and per-email storage key	~20
`mime_utils/email_builder.py`	Constructs RFC 2822 MIME messages with headers, body, and attachments	
`mime_utils/email_parser.py`	Parses raw MIME bytes into structured data	
`mime_utils/attachment_handler.py`	Safely extracts and saves attachments	
`kem/server.py`	Server-side KEM handshake (decapsulation)	
`kem/client.py`	Client-side KEM handshake (encapsulation)	
`mc_eliece_lib/core.py`	Calls compiled C binaries (`keygen`, `find_H`, `encryption`, `decryption`)	
`mc_eliece_lib/utils.py`	Matrix parsing; HKDF secret derivation	

3.3 On-Disk Data Layout

```
mailbox/
  <user>/
    inbox/
      <id>.json    # metadata: from, subject, date, read flag
      <id>.enc     # AES-256-CBC encrypted MIME message
      <id>.key     # per-email storage AES key
received_files/
  <user>/          # extracted attachments after receive
temp_files/        # KEM intermediate matrix files (transient)
```

Listing 1: Mailbox directory structure

4 Cryptographic Design

4.1 Three-Layer Encryption Model

Each email passes through three distinct encryption layers, illustrated in Figure 2.

1. **Transport layer (sender → server):** AES-256-CBC under session key S_1 derived from the sender's KEM handshake.
2. **Storage layer (at rest on server):** AES-256-CBC under a fresh random per-email key K generated by the server (`os.urandom(32)`).
3. **Transport layer (server → receiver):** K itself is re-encrypted under the receiver's session key S_2; the encrypted email content (under K) is transmitted alongside.

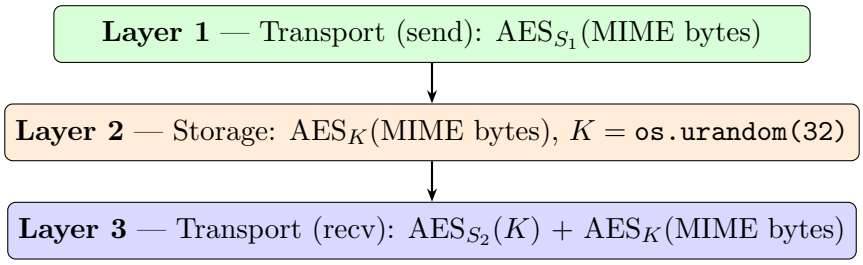

Figure 2: Three-layer encryption model for one email lifecycle.

4.2 KEM Handshake Protocol

The handshake is performed for every client connection and proceeds as follows:

1. Server creates a `KEMServer` instance and calls `perform_handshake(conn)`.
2. Server generates a McEliece key pair and transmits the public matrix $G_{\bar{\omega}}$ to the client as a JSON-framed message.
3. Client calls `KEMClient.perform_handshake(sock)`, receives the public key, and runs encapsulation:
$$c = m_\omega \cdot G_{\bar{\omega}} + e, \quad \|e\|_0 \leq t$$
where m_ω is a randomly chosen message and e is a random error vector of weight at most t.

4. Client derives the session key via HKDF:

$$S = \text{HKDF}(m_\omega, \text{salt}, \text{info})[0:32]$$

and sends ciphertext c to the server.

5. Server decapsulates c using its private key to recover m_ω, derives the same S via HKDF.

6. Both parties now share S and use it as an AES-256-CBC key for the remainder of the connection.

4.3 Forward Secrecy

Because a new key pair is generated for each connection, compromise of a stored key does not expose any previous session. This property — forward secrecy — is critical for the harvest-now-decrypt-later threat model: even if an adversary records all ciphertext and later obtains the server's long-term state, past session keys are irrecoverable.

4.4 Security Properties Summary

Table 2: Security properties of the system

Property	**Mechanism**	**Status**
Quantum-resistant key exchange	McEliece KEM	✓
Bulk data confidentiality	AES-256-CBC	✓
Forward secrecy	Ephemeral KEM keys per connection	✓
Storage isolation	Per-email random key K	✓
Eavesdropping resistance	End-to-end AES encryption	✓
Harvest-now-decrypt-later	Post-quantum KEM	✓
Endpoint compromise	Out of scope (trusted endpoints)	×
Metadata hiding	Not implemented	×
Server admin access	Server is trusted	×

5 Protocol Flow

5.1 Server Startup

1. Execute `python3 mime_server_with_kem.py`.

2. `MIMEServerWithKEM` is instantiated: initialises `MailboxManager`, `AttachmentHandler`, `EmailParser`, and reads KEM parameters from environment variables (`KEM_Q`, `KEM_K`, `KEM_T`).

3. Server binds TCP socket on `0.0.0.0:8025` (default) and enters accept loop.

4. Each accepted connection is dispatched to a new thread running `handle_client(...)`.

5.2 SEND Flow (Alice → Server)

1. Client builds a MIME message (From, To, Subject, body, attachments).
2. Client connects and completes the KEM handshake, obtaining session key S_1.
3. Client constructs a payload dict {`recipient, mime_bytes`}, pickles it, and encrypts with AES-256-CBC under S_1.
4. Client sends: command string `SEND`, 16-byte size field, encrypted payload.
5. Server decrypts payload with S_1, unpickles, and parses MIME.
6. Server generates storage key $K =$ `os.urandom(32)`.
7. Server re-encrypts MIME bytes under K.
8. Server calls `MailboxManager.store_email(...)`: writes `<id>.json`, `<id>.enc`, `<id>.key`.
9. Server saves extracted attachments to `received_files/<recipient>/`.
10. Server returns `OK` acknowledgement.

5.3 RECV Flow (Server → Bob)

1. Client connects and completes a fresh KEM handshake, obtaining session key S_2.
2. Client sends command `RECV <username>`.
3. Server lists email metadata in the user's inbox. If empty, sends integer `0` and closes.
4. For each email the server builds a *bundle*:
   - Encrypted content $\text{AES}_K(\text{MIME})$
   - Metadata JSON
   - Storage key re-encrypted: $\text{AES}_{S_2}(K)$
5. Server pickles all bundles, sends size + bundle bytes.
6. Client receives bundles; for each:
   a. Decrypts $\text{AES}_{S_2}(K)$ to recover K.
   b. Decrypts $\text{AES}_K(\text{MIME})$ to obtain plaintext MIME bytes.
   c. Parses MIME; prints From/To/Date/Subject/body.
   d. Saves attachments locally.
7. Server marks each retrieved email as `read=true` in its metadata JSON.

5.4 LIST Flow (Headers Only)

1. Client connects, handshakes, sends `LIST <username>`.
2. Server builds a lightweight list: {`id, from, subject, date, read, has_attachment`}.
3. Server pickles list, encrypts under S, sends size + data.
4. Client decrypts and prints inbox summary.

6 Implementation Details

6.1 McEliece Library (`mc_eliece_lib`)

The core cryptographic operations are offloaded to compiled C binaries located in `mc_eliece_lib/binari`

`keygen`
: Generates a random Goppa code over $\mathbb{F}_q$ with dimension k and error-correction capacity t. Produces the generator matrix G and its systematic form.

`find_H`
: Computes the dual (parity-check) matrix H, needed for syndrome-based decoding.

`encryption`
: Encapsulation: given public matrix $G_{\bar{\omega}}$ and message m_ω, produces ciphertext c.

`decryption`
: Decapsulation: given private parameters and ciphertext c, recovers m_ω via Goppa decoding.

`mc_eliece_lib/utils.py` handles matrix file I/O and HKDF secret derivation (`derive_secret_from`

6.2 KEM Parameters and Security Levels

The system is parameterised by three values (q, k, t), where q is the field size, k is the code dimension, and t is the error-correction capacity. Table 3 lists the available presets.

Table 3: McEliece KEM parameter sets and performance

q	k	t	**Security**	**Handshake**	**Use Case**
3	22	8	Low	∼1.6 s	Testing only
5	45	15	Medium	∼1.8 s	**Recommended**
7	172	60	High	∼7.2 s	Balanced
11	666	100	Very High	∼8.3 min	Maximum practical
13	1099	120	Maximum	∼36 min	Research / offline

Parameters are set via environment variables before launching either the server or the client, ensuring both sides operate at the same security level.

6.3 MIME Message Structure

The `email_builder` module produces RFC-compliant MIME messages:

```
From: alice@localhost
To: bob@localhost
Subject: Quantum-Safe Email
Date: Wed, 28 Feb 2026 14:32:15 +0000
Message-ID: <uuid@localhost>
MIME-Version: 1.0
Content-Type: multipart/mixed; boundary="BOUNDARY"

--BOUNDARY
Content-Type: text/plain; charset=utf-8

This message is protected against quantum attacks!

--BOUNDARY
Content-Type: application/pdf
Content-Disposition: attachment; filename="report.pdf"
Content-Transfer-Encoding: base64

[base64 encoded data]
--BOUNDARY--
```

Listing 2: Example MIME message with attachment

6.4 Serialisation and Transport Protocol

All structured payloads (email bundles, metadata lists) are serialised with Python's `pickle` and encrypted before transmission. The wire format for each command is:

Command string	16-byte size field	Encrypted payload bytes

This minimalist framing avoids implementing a full SMTP dialogue while remaining extensible.

7 Performance Analysis

7.1 Handshake Overhead

The dominant cost in the system is the McEliece KEM handshake, specifically the key generation and Goppa decoding steps performed by the compiled C binaries. Email payload encryption (AES-256-CBC) is negligible by comparison: a 1 MB email transfers in approximately 50 ms regardless of the parameter set.

Figure 3 illustrates the trade-off between security level and handshake latency.

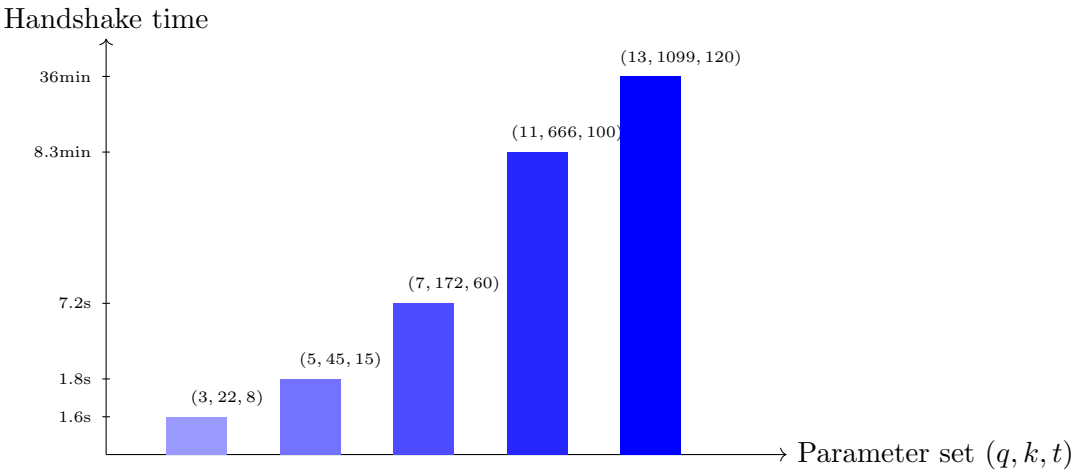

Figure 3: McEliece KEM handshake time vs. parameter set (not to linear scale). The $(5, 45, 15)$ preset provides the best usability–security trade-off.

7.2 Recommended Deployment Configuration

For production use, the $(q = 5, k = 45, t = 15)$ preset is recommended:

- Handshake: $\approx 1.8\,\text{s}$
- Key size: $\approx 45\,\text{KB}$
- Email transfer: $\approx 50\,\text{ms}$ per MB
- Total send operation: $\approx 2\,\text{s}$ end-to-end
- Post-quantum security level: 128-bit

8 Security Analysis and Threat Model

8.1 Threats Mitigated

Quantum attacks on key exchange
The McEliece KEM is conjectured to be secure against both classical and quantum adversaries. The best known quantum attack (information-set decoding variants) achieves only a polynomial speedup, far below the exponential advantage Shor's algorithm provides against RSA.

Man-in-the-middle attacks
The KEM handshake binds session keys to the specific connection; any tampering with the ciphertext produces an incorrect decapsulation and no shared secret.

Eavesdropping
All data is encrypted with AES-256-CBC before transmission.

Harvest-now-decrypt-later
Because both the transport keys (McEliece KEM) and the storage keys (random AES) are quantum-resistant, recorded ciphertext cannot be decrypted by a future quantum computer.

8.2 Limitations and Out-of-Scope Threats

Compromised endpoints
If a client machine is infected by malware, plaintext messages are exposed before encryption. This is outside the scope of the cryptographic design.

Server administrator access
The server processes plaintext MIME during the SEND operation (between decryption and re-encryption with the storage key). A malicious administrator could intercept messages at this point. Fully end-to-end encrypted email would require the server never to see plaintext, which would require the receiver's public key at send time.

Traffic analysis / metadata
Sender and receiver IP addresses, timestamps, and message sizes are observable in transit. Anonymous routing (e.g. Tor) is not included.

Replay attacks
The current protocol does not include nonces or timestamps in the KEM handshake to prevent replay. This should be addressed in a production hardening pass.

9 Usage Guide

9.1 Installation and Dependencies

```
# Python dependency
pip install cryptography

# Compile McEliece C binaries
cd mc_eliece_lib
make

# Python >= 3.6 required; ~100 MB disk for mailboxes
```

Listing 3: Setup

9.2 Running the System

```
# Terminal 1     start server (default parameters)
python3 mime_server_with_kem.py

# Terminal 2     Alice sends an email with attachment
python3 mime_client_with_kem.py send \
    --from alice@localhost \
    --to   bob@localhost \
    --subject "Quantum-Safe␣Hello" \
    --body "Protected␣by␣McEliece␣KEM!" \
    --attach ./report.pdf

# Terminal 3     Bob lists inbox headers
```

```
python3 mime_client_with_kem.py list --user bob

# Terminal 3     Bob downloads full emails
python3 mime_client_with_kem.py receive --user bob
```

Listing 4: Basic usage

```
# High-security mode (Q=7)
KEM_Q=7 KEM_K=172 KEM_T=60 python3 mime_server_with_kem.py
KEM_Q=7 KEM_K=172 KEM_T=60 python3 mime_client_with_kem.py send
   ...
```

Listing 5: Custom security parameters

10 Future Work

Several enhancements would strengthen the system for production deployment:

1. **Full end-to-end encryption:** Cache recipient public keys so the sender can encrypt directly to the recipient, eliminating the server's brief access to plaintext.
2. **IMAP-style folder management:** Drafts, Sent, Trash, and labelling for richer UX.
3. **Multi-recipient support:** To, CC, and BCC fields, each requiring separate KEM operations or a multi-recipient KEM scheme.
4. **Replay protection:** Include per-handshake nonces and timestamps.
5. **TLS wrapping:** Optionally layer a post-quantum TLS connection (e.g. using a Kyber-based TLS stack) beneath the application protocol for defence in depth.
6. **Message threading:** In-reply-to / References headers for conversation view.
7. **NIST-standardised McEliece:** Migrate to the NIST-standardised Classic McEliece parameter sets once finalised.

11 Conclusion

This project demonstrates a fully functional, quantum-safe asynchronous email system built on the McEliece Key Encapsulation Mechanism. By replacing the RSA-based key exchange traditionally used in S/MIME and PGP with a code-based post-quantum alternative, the system achieves resistance to both present-day and anticipated quantum attacks.

The three-layer encryption model — ephemeral session key for transport, random per-email storage key, and session-key-wrapped delivery — provides forward secrecy and storage isolation without requiring the sender and receiver to be online simultaneously. Full MIME compliance, including multi-part messages and file attachments, is maintained throughout.

Performance benchmarks show that the recommended $(q = 5, k = 45, t = 15)$ parameter set incurs only a $\sim$1.8 s handshake overhead per send or receive operation, making

the system practical for interactive use while providing 128-bit post-quantum security. Higher parameter sets are available for applications requiring stronger guarantees, at the cost of increased latency.

The system stands as a proof-of-concept for the viability of post-quantum email and a concrete reference implementation for integrating the McEliece cryptosystem into application-layer protocols.

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